

Sales Prediction Using Machine Learning

Sales prediction is the process of estimating future sales based on historical data. In this project, machine learning techniques are used to predict product sales based on advertising expenditure across different platforms.

Objective

The main objective of this project is to build a regression model that can predict sales based on TV, Radio, and Newspaper advertising budgets.

Dataset Description

The dataset contains advertising spending data and corresponding sales figures. It includes four columns: TV, Radio, Newspaper, and Sales.

Methodology

1. Loaded dataset from a CSV file.
2. Performed exploratory data analysis to understand data patterns.
3. Selected relevant features for prediction.
4. Trained a Linear Regression model.
5. Evaluated the model using MSE and R^2 score.

Results

The Linear Regression model achieved good performance with a high R^2 score, indicating a strong relationship between advertising expenditure and sales.

Conclusion

This project demonstrates how machine learning can be effectively used to predict sales. Such predictions help businesses optimize advertising strategies and maximize revenue.